CYBER SECURITY AWARENESS AND PHISHING DETECTION TOOLKIT

MAJOR PROJECT REPORT

RAJA NAYAK

Cybersecurity

Duration: 14 Days

Mode: Individual Project

Organization: Quillance Infotech Pvt. Ltd.

Submission: Report + Source Files + Presentation + Demo Video

2026

STUDENT DECLARATION

I hereby declare that the project titled “Cyber Security Awareness & Phishing Detection Toolkit” has been completed by me as an individual academic project. I have prepared the project documentation, source files, testing activities, and supporting materials as part of my own work. I understand the project requirements and take responsibility for the accuracy and authenticity of the submitted work.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Quillance Infotech Pvt. Ltd. for providing me with the opportunity to work on this academic project. I am thankful for the guidance, learning resources, and support that helped me understand phishing awareness, risk assessment, and basic cybersecurity concepts. I also appreciate everyone who contributed to the successful completion of this project.

ABSTRACT

This project presents a practical Cyber Security Awareness & Phishing Detection Toolkit designed to improve understanding of phishing threats and help users identify common warning signs. The toolkit combines phishing knowledge, an indicator library, rule-based risk scoring, a Python checker, testing activities, and a cybersecurity awareness checklist. The project focuses on safe educational examples and demonstrates how indicators such as urgency, credential requests, suspicious links, and unusual sender wording can be used to assess phishing risk. The expected outcome is a simple educational toolkit that helps users develop safer habits when dealing with suspicious digital communications.

TABLE OF CONTENTS

Contents

CYBER SECURITY AWARENESS AND PHISHING DETECTION TOOLKIT 1

STUDENT DECLARATION 2

ACKNOWLEDGEMENT 3

ABSTRACT 4

TABLE OF CONTENTS 5

1. INTRODUCTION AND BACKGROUND 8

1.1 Project Background 8

1.2 Project Purpose 8

1.3 Project Scope 8

1.4 Project Objectives 8

1.5 Project Modules 8

1.6 Expected Outcome 9

2. PROBLEM STATEMENT 10

2.1 Problem Content 10

2.2 Proposed Problem Approach 10

2.3 Project Requirements 10

2.4 Tools and Technologies 10

2.5 Methodology 10

2.6 Project Workflow 11

3. SYSTEM / DATA PREPARATION / PROCESS DESIGN 12

3.1 System Design 12

3.2 Data Preparation 12

3.3 Process Design 12

3.4 System Components 12

3.5 Risk Scoring Model 12

3.6 Risk Classification 13

3.7 Detection Indicators 13

3.8 System Flow 13

3.9 Data Preparation 13

4. IMPLEMENTATION / ANALYSIS 14
4.1 Python Checker Implementation 14
4.2 Indicator Detection Logic 14
4.3 Risk Score Calculation 14
4.4 Risk Classification Logic 14
4.5 Output and User Guidance 14
4.6 Indicator and Rule File 14
4.7 Testing Implementation 15
4.8 Implementation Limitations 15
5. TESTING AND RESULTS 16
5.1 Testing Environment 16
5.2 Test Cases 16
5.3 Testin Results Summary 17
5.4 Testing Results Analysis 17
5.5 Screenshots of Application Output 17
5.6 Testing Observations 19
5.7 False Positives and False Negatives 19
5.8 Testing Limitations 19
6. SCREENSHOTS, TABLES AND CHARTS 20
6.1 Application Output Screenshots 20
6.2 Testing Results Table 21
6.3 Risk Classification Chart 21
6.4 Indicator Detection Summary 22
6.5 Visual Evidence 22
7. KEY FINDINGS 23
7.1 Key Findings 23
7.2 Overall Findings 23
8. CHALLENGES AND LIMITATIONS 24
8.1 Challenges 24
8.2 Project Limitations 24
8.3 Safety and Ethical Considerations 24
9. FUTURE SCOPE / IMPROVEMENTS 25

9.1 Future Improvements 25
9.2 Scalability and Environment 25
10. CONCLUSION 26
11. REFERENCES.....27

1. INTRODUCTION AND BACKGROUND

Phishing is a common social engineering threat in which attackers use deceptive messages and other communication methods to trick users into revealing sensitive information or performing unsafe actions. Phishing attempts may use urgency, suspicious links, credential requests, unusual sender wording, or account-related warnings to influence users. This project focuses on creating a safe educational toolkit that helps users understand these warning signs and assess phishing risk using simple, explainable rules.

1.1 Project Background

Phishing is a common social engineering technique that uses deceptive communication to influence users into taking unsafe actions. These attacks may involve urgency, suspicious links, credential requests, account warnings, or unusual sender wording. The project brief identifies these types of indicators as important elements for phishing awareness and risk assessment.

1.2 Project Purpose

The purpose of this project is to develop a practical and manageable cybersecurity awareness toolkit that helps users understand phishing threats and identify common warning indicators. The toolkit is designed as an educational project using safe sample data and simple rule-based analysis. It aims to demonstrate phishing awareness, risk scoring, and safe user practices without unnecessary advanced complexity.

1.3 Project Scope

The scope of the project covers the preparation of phishing awareness content, a library of common phishing indicators, a simple rule-based risk scoring model, a Python-based checker, safe sample messages for testing, and an awareness checklist. The project also includes testing, recording results, documenting limitations, and preparing the final report, presentation, and demonstration. The project is limited to safe, synthetic, or publicly available training examples and does not involve real-world phishing activities.

1.4 Project Objectives

- Understand phishing and common social engineering techniques.
- Design a simple rule-based phishing risk scoring model.
- Build a Python program that analyzes pasted sample email text.
- Classify messages as Low, Medium or High Risk.
- Create a user-friendly security awareness checklist.

1.5 Project Modules

1. Phishing Knowledge Base – Documents phishing, spear phishing, fake login pages, suspicious attachments, spoofed sender patterns, and social engineering.
2. Indicator Library – Contains common warning indicators such as urgency language, credential requests, prizes, account suspension, suspicious URLs, and unusual sender wording.
3. Risk Scoring – Assigns points to detected indicators and defines Low, Medium, and High risk thresholds.

4. Python Checker – Accepts sample email text, detects warning indicators, calculates the risk score, and displays reasons for the classification.
5. Testing – Tests safe sample messages, including legitimate-style and phishing-style examples, and records the results.
6. Awareness Guide – Provides a practical checklist explaining what users should verify before clicking links or sharing information.

1.6 Expected Outcome

The expected outcome of the project is a working phishing awareness and detection toolkit supported by clear documentation and practical testing. The project should demonstrate the use of phishing indicators, rule-based risk scoring, sample message analysis, and cybersecurity awareness guidance. The completed work should also enable the student to explain the project approach, implementation, important results, and limitations during evaluation.

2. PROBLEM STATEMENT

Phishing messages often use urgency, suspicious links, credential requests, and other deceptive techniques to trick users into taking unsafe actions. Users may find it difficult to distinguish legitimate communications from phishing-style messages, especially when attackers make the communication appear trustworthy. This creates a need for a simple educational toolkit that can help users understand common phishing indicators and assess the risk of sample messages using explainable rules.

2.1 Problem Content

The project brief identifies phishing messages as a practical cybersecurity problem because they can use urgency, suspicious links, credential requests, and deceptive communication to influence users. It also emphasizes that the project should remain safe and educational, using synthetic or publicly available training examples rather than real-world phishing activity.

2.2 Proposed Problem Approach

The project addresses the problem by creating a safe educational toolkit that uses a phishing knowledge base, an indicator library, and a simple rule-based risk scoring model. A Python checker is used to analyze pasted sample email text, detect warning indicators, calculate a risk score, and classify the message as Low, Medium, or High Risk. The toolkit also provides awareness guidance to help users understand and respond to suspicious messages safely.

2.3 Project Requirements

The project requires Python 3 for development of the phishing checker, with Visual Studio Code or PyCharm as the development environment. CSV or text files can be used for sample and rule data, while Microsoft PowerPoint is required for the final presentation. The project also requires a Python source file, an indicator/rule file, a 20-message test sheet with expected and actual results, screenshots of application output, a cybersecurity awareness checklist, a final PDF report, an 8–12 slide presentation, and a 3–5 minute demo video.

2.4 Tools and Technologies

The project uses Python 3 as the primary programming language for developing the phishing detection checker. Visual Studio Code is used as the development environment for writing and executing the Python program. CSV or text files can be used to store sample messages and phishing indicators. Microsoft PowerPoint is used to prepare the project presentation. These tools provide a simple and suitable environment for developing, testing, and documenting the educational phishing detection toolkit.

2.5 Methodology

The project follows a structured methodology that begins with understanding phishing threats and common social engineering techniques. Safe sample messages and phishing indicators are then prepared for analysis. A simple rule-based scoring approach is designed to assign points to detected indicators and classify messages as Low, Medium, or High Risk. The Python checker is developed to analyze sample email text, identify warning indicators, calculate the risk score, and display the reasons for the result. The tool is tested using safe sample messages, and the results are recorded to evaluate

its effectiveness and identify limitations. Awareness guidance is also prepared to help users respond safely to suspicious communications.

2.6 Project Workflow

The project workflow follows a step-by-step process. First, phishing concepts and common social engineering techniques are studied. Next, safe sample messages and phishing indicators are prepared. A rule-based risk scoring model is then designed, followed by development of the Python checker for sample email text analysis. The checker identifies warning indicators, calculates a risk score, and provides a risk classification with reasons. The tool is then tested using safe sample messages, and the results are recorded and reviewed. Finally, the awareness checklist, documentation, presentation, and demonstration materials are prepared for submission.

3. SYSTEM / DATA PREPARATION / PROCESS DESIGN

3.1 System Design

The system is designed as a simple educational phishing detection workflow. The user provides a sample email message to the Python checker. The checker examines the text for predefined phishing indicators, such as urgent language, requests for credentials, suspicious links, account warnings, and unusual sender wording. Each detected indicator contributes to the overall risk score. The final score is used to classify the message as Low, Medium, or High Risk, and the detected indicators are displayed as reasons for the result.

3.2 Data Preparation

The project uses safe and educational sample data for testing and analysis. The data preparation process includes creating sample phishing-style and legitimate-style messages and identifying common warning indicators within them. The prepared data is used to test the rule-based detection logic and evaluate whether the expected risk classification matches the actual result produced by the Python checker. No real passwords, private credentials, confidential company information, or real-world phishing activity are used.

3.3 Process Design

The process begins when a user provides a safe sample email message to the phishing detection checker. The system analyzes the message against the predefined indicator rules. Detected indicators are assigned corresponding risk points, and the points are combined to calculate the overall risk score. Based on the defined scoring thresholds, the message is classified as Low, Medium, or High Risk. The checker then displays the risk level, score, detected indicators, and safety recommendations to help the user understand the result.

3.4 System Components

The toolkit consists of several connected components that work together to support phishing awareness and detection. The main components include the Phishing Knowledge Base, Indicator Library, Risk Scoring Module, Python Checker, Testing Module, and Awareness Guide. The knowledge base provides basic phishing information, while the indicator library contains predefined warning signs. The risk scoring module assigns points to detected indicators, and the Python checker analyzes sample email text and displays the result. The testing module records expected and actual results, while the awareness guide provides practical safety recommendations.

3.5 Risk Scoring Model

The risk scoring model uses predefined phishing indicators to assign risk points to a sample message. Indicators such as urgency, credential requests, suspicious links, account warnings, and unusual sender wording contribute to the overall score. The total score is then compared with predefined thresholds to classify the message as Low, Medium, or High Risk. This rule-based approach makes the analysis simple, explainable, and suitable for an educational cybersecurity project.

3.6 Risk Classification

The toolkit classifies analyzed sample messages into three risk levels based on the total risk score. A Low Risk result indicates that few or no predefined phishing indicators were detected. A Medium Risk result indicates that multiple warning indicators were identified and the message should be reviewed carefully. A High Risk result indicates a higher concentration of suspicious indicators and requires greater caution. The classification is intended for educational awareness and does not guarantee that a message is safe or malicious.

3.7 Detection Indicators

The toolkit uses predefined indicators to identify characteristics commonly associated with phishing messages. Important indicators include urgent or threatening language, requests for credentials or sensitive information, suspicious links, account suspension or verification messages, unexpected attachments, unusual sender details, and social engineering patterns. These indicators are checked by the rule-based analysis process and contribute to the overall risk assessment.

3.8 System Flow

The system flow follows a simple sequence from input to final risk classification. The user first provides a safe sample email message to the Python checker. The message is then examined for predefined phishing indicators. Detected indicators are assigned risk points, and the points are combined to calculate the total risk score. The score is compared with the defined risk thresholds, after which the system displays the risk level, detected indicators, and safety recommendations.

3.9 Data Preparation

The testing dataset is prepared using safe, synthetic sample messages representing both legitimate-style and phishing-style communications. Each sample is assigned an expected risk classification based on the predefined indicators and scoring rules. During testing, the Python checker analyzes each message and produces an actual risk level and score. The expected and actual results are recorded for comparison so that the detection logic can be evaluated and improved.

4. IMPLEMENTATION / ANALYSIS

4.1 Python Checker Implementation

The Python checker is the main analysis component of the toolkit. It is designed to accept a sample email message as input and examine the content using predefined phishing indicators and rules. The checker calculates a risk score based on the detected indicators and assigns a Low, Medium, or High Risk classification. It also displays the detected indicators and provides understandable reasons for the generated result. The implementation is intended for safe educational testing using synthetic sample messages.

4.2 Indicator Detection Logic

The indicator detection logic checks the sample email text for predefined warning signs associated with phishing. The rules can identify patterns such as urgent or threatening language, requests for credentials or sensitive information, suspicious links, account verification or suspension messages, unexpected attachments, and unusual sender wording. When an indicator is detected, the corresponding rule contributes to the risk score and the detected reason is shown to the user. This makes the analysis transparent and easier to understand.

4.3 Risk Score Calculation

The risk score is calculated by adding the points assigned to each detected phishing indicator. Indicators with a higher level of concern can contribute more points to the total score. After all applicable rules are checked, the individual points are combined to produce the final risk score. The score is then compared with the predefined thresholds to determine whether the sample message is classified as Low, Medium, or High Risk.

4.4 Risk Classification Logic

The risk classification logic converts the calculated score into an understandable risk level. A lower score represents Low Risk, a moderate score represents Medium Risk, and a higher score represents High Risk. The classification helps users quickly understand the level of caution required when reviewing a sample message. The result is presented together with the detected indicators so that the user can understand why a particular risk level was assigned.

4.5 Output and User Guidance

The checker presents the analysis result in a clear and understandable format. The output includes the analyzed message information, calculated risk score, risk classification, and the phishing indicators detected by the rules. The system also provides basic safety guidance, such as avoiding suspicious links, not sharing sensitive information, and verifying unexpected requests through trusted channels. This helps users understand both the result and the safe action they should take.

4.6 Indicator and Rule File

The indicator and rule file contains the predefined patterns used by the phishing detection checker. These rules represent common phishing warning signs such as urgency, credential requests, suspicious links, account-related warnings, and unusual sender wording. Keeping the indicators and rules

organized separately makes the detection logic easier to review, update, and test. The rule file is used together with the Python checker to support consistent risk assessment of safe sample messages.

4.7 Testing Implementation

The implemented checker is tested using safe sample messages prepared for educational purposes. Testing includes both legitimate-style and phishing-style examples so that the detection rules can be evaluated under different conditions. For each test message, the expected risk level is compared with the actual result produced by the checker. The test results are recorded to identify incorrect classifications, review the effectiveness of the rules, and improve the scoring logic where necessary.

4.8 Implementation Limitations

The current implementation uses a simple rule-based approach and is intended for educational purposes. It does not provide real-time threat intelligence, machine learning-based detection, external domain reputation checking, or access to live phishing databases. Therefore, the checker may produce false positives or fail to identify advanced phishing techniques. The results should be treated as an awareness aid rather than a guarantee that a message is safe or malicious.

5. TESTING AND RESULTS

5.1 Testing Environment

The phishing detection toolkit is tested in a Python development environment using Visual Studio Code. Testing is performed with safe, synthetic sample messages representing legitimate-style and phishing-style communications. Each message is analyzed by the checker using the predefined indicator and scoring rules. The generated risk score, risk classification, and detected indicators are recorded for comparison with the expected results.

5.2 Test Cases

Test Case 1 – Legitimate-Style Message

Sample Message:

From: support@example.test
Subject: Scheduled System Maintenance

Dear User,
Our system will undergo scheduled maintenance on Sunday. You may experience temporary service interruptions during this period. No action is required from you.

Thank you,
Support Team

Expected Result: LOW RISK

Actual Result: LOW RISK

Risk Score: 0

Test Case 2 – Phishing-Style Message

Sample Message:

From: security-alert@example.test
Subject: Urgent: Verify Your Account

Dear User,
We detected unusual activity on your account. Please verify your account immediately to avoid temporary suspension. Click the link provided in the message to complete the verification process.

Thank you,
Security Team

Expected Result: MEDIUM RISK

Actual Result: MEDIUM RISK

Risk Sore: 3

Test Case 3 – Multiple Phishing Indicators

Sample Message:

From: account-alert@example.test
Subject: Urgent: Verify Your Account

Dear User,
Your account has been flagged due to suspicious activity. Please verify your login details and password immediately to prevent account suspension. Follow the verification instructions provided in this message.

Thank you,
Account Security Team

Expected Result: HIGH RISK

Actual Result: HIGH RISK

Risk Score: 6

5.3 Testin Results Summary

Test Case	Message Type	Expected Result	Actual Result
Test 1	Legitimate-Style Message	Low Risk	Low Risk
Test 2	Phishing-Style Message	Medium Risk	Medium Risk
Test 3	Multiple Phishing Indicators	High Risk	High Risk

A total of 20 safe sample messages were tested using the phishing detection tool. The test set included legitimate-style and phishing-style messages to evaluate the tool's risk classification capability. The testing results recorded 5 Low Risk, 11 Medium Risk, and 4 High Risk classifications. Detailed test messages, expected results, actual results, risk scores, and detected indicators are provided in the accompanying 20-message testing sheet.

5.4 Testing Results Analysis

A total of 20 safe sample messages were tested using the phishing detection tool. The test set included legitimate-style and phishing-style messages to evaluate the tool's risk classification capability. The testing results recorded 5 Low Risk, 11 Medium Risk, and 4 High Risk classifications. Detailed test messages, expected results, actual results, risk scores, and detected indicators are provided in the accompanying 20-message testing sheet.

5.5 Screenshots of Application Output

Screenshots of the phishing detection checker are included to demonstrate the application output during testing. The screenshots show the analyzed input, calculated risk score, risk classification, and detected phishing indicators. These outputs provide visual evidence of the implemented detection and scoring process.

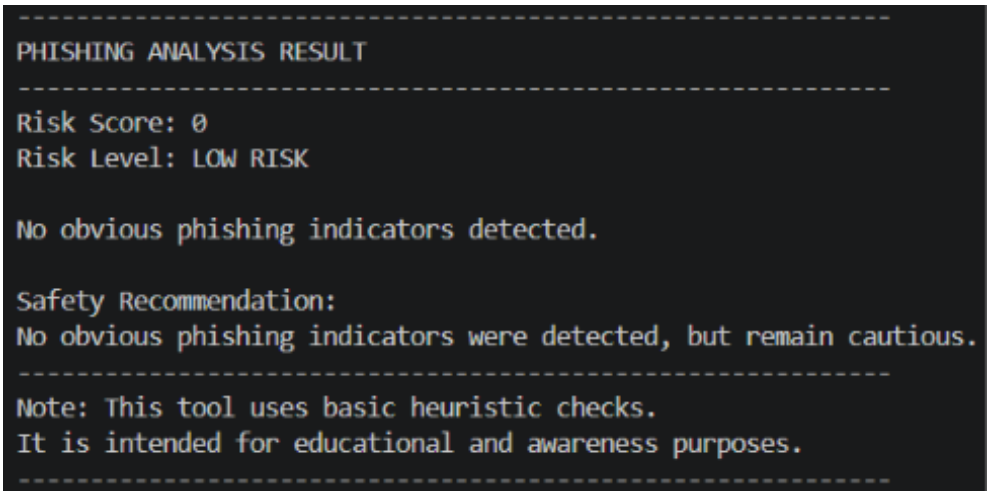


Figure 4: Email Analysis – Low Risk Classification

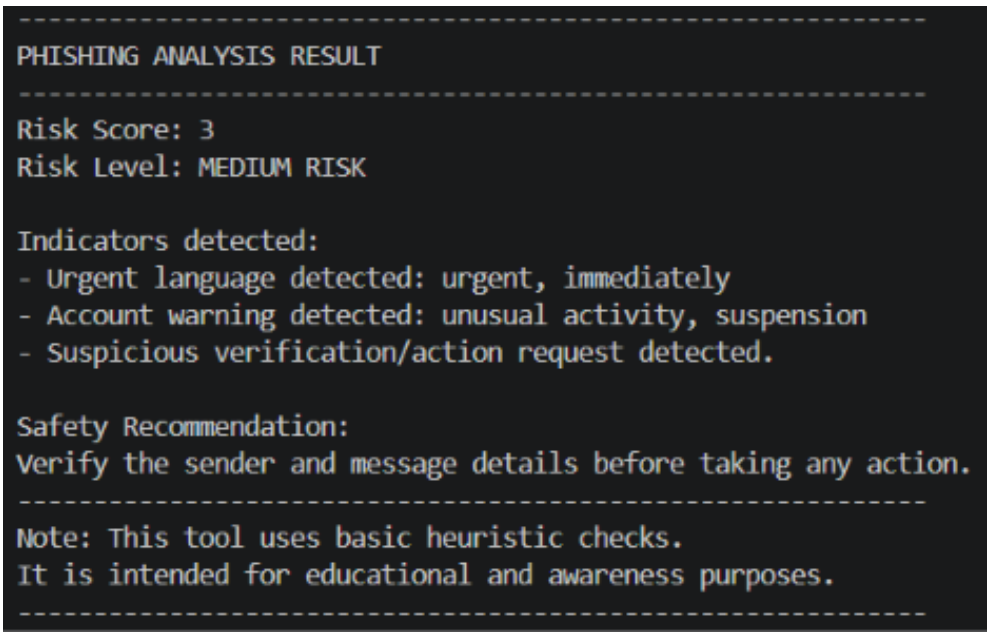


Figure 5: Email Analysis – Medium Risk Classification

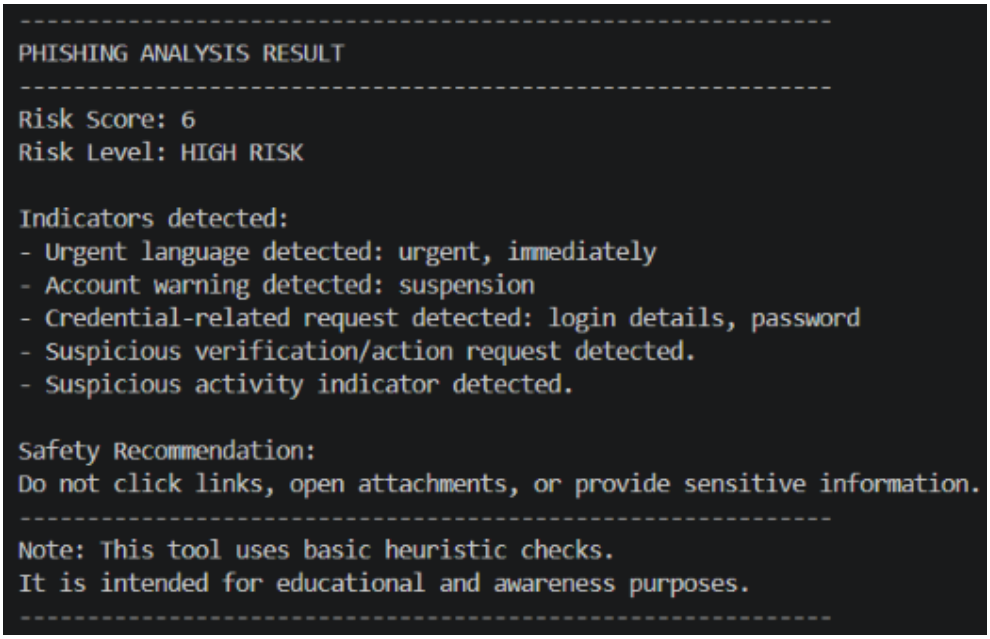


Figure 6: Email Analysis – High Risk Classification

5.6 Testing Observations

During testing, the checker successfully identified the predefined phishing indicators present in the sample messages and produced different risk classifications based on the calculated scores. The legitimate-style message received a Low Risk result, while messages containing suspicious characteristics received higher risk classifications. The observations show that combining multiple indicators can increase the risk score and provide a clearer warning to the user. These results also highlight the importance of reviewing the detected indicators rather than relying only on the final risk level.

5.7 False Positives and False Negatives

Because the toolkit uses predefined rules and indicators, incorrect classifications are possible. A false positive may occur when a legitimate message contains words or patterns that resemble phishing indicators. A false negative may occur when a phishing-style message does not contain any of the predefined indicators. Reviewing these cases is important for improving the indicator library and scoring rules. The current testing provides a basic evaluation of the rule-based approach using safe sample messages.

5.8 Testing Limitations

The testing is limited to a small set of safe, synthetic sample messages and predefined phishing indicators. It does not represent the full range of real-world phishing techniques or attacker behavior. The results may also vary depending on the wording and structure of a message. Therefore, the testing demonstrates the educational value of the toolkit but should not be considered a complete measure of real-world phishing detection accuracy.

6. SCREENSHOTS, TABLES AND CHARTS

6.1 Application Output Screenshots

The following screenshots provide visual evidence of the phishing detection checker during testing. They demonstrate the input provided to the tool, the calculated risk score, the resulting risk classification, and the indicators detected by the system.

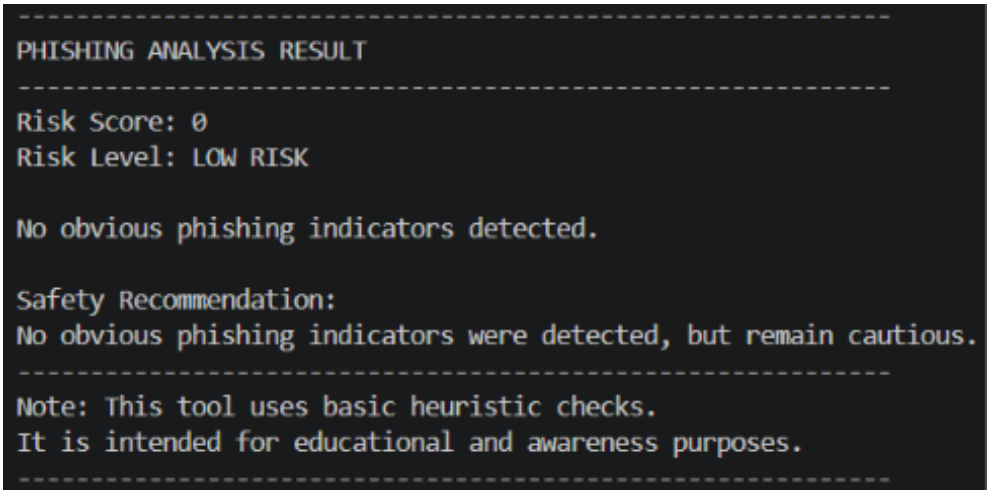


Figure 1: Email Analysis - Low Risk Classification

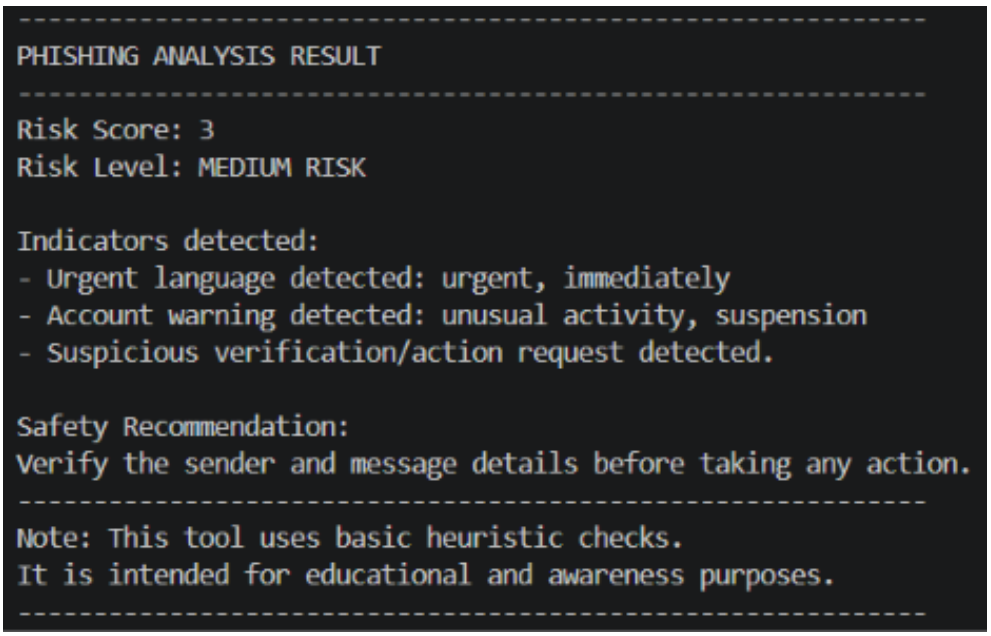


Figure 2: Email Analysis – Medium Risk Classification

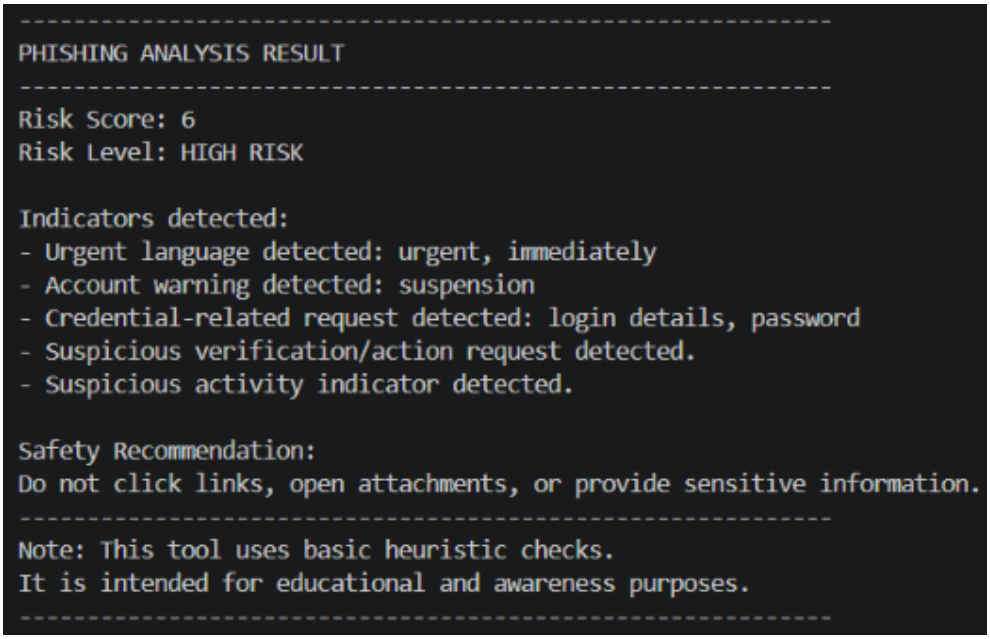


Figure 3: Email Analysis – High Risk Classification

6.2 Testing Results Table

The testing results are summarized in the table below to provide a clear comparison between the expected and actual classifications obtained from the phishing detection checker. The table helps demonstrate whether the implemented rules produced the intended risk levels for the selected safe sample messages.

Test Case	URL Type	Risk Score	Risk Level
Test 1	Legitimate – Style Message	0	Low Risk
Test 2	Phishing-Style Message	3	Medium Risk
Test 3	Multiple Phishing Indicators	6	High Risk

6.3 Risk Classification Chart

The risk classification chart provides a visual representation of the risk levels obtained during testing. It helps compare the number of sample messages classified as Low, Medium, and High Risk. The chart will be prepared using the recorded testing results to make the overall results easier to understand.



6.4 Indicator Detection Summary

The indicator detection summary presents the main phishing warning signs identified during the testing process. These indicators include urgency, credential requests, account-related warnings, suspicious links, and unusual sender wording. Reviewing the detected indicators helps explain why a particular risk score and classification were assigned to each sample message.

6.5 Visual Evidence

The project includes screenshots and documented testing results as visual evidence of the implemented toolkit. The evidence demonstrates the working process of the checker and supports the recorded test results. Screenshots are included to show how different sample messages are analyzed and how the system presents the corresponding risk assessment.

7. KEY FINDINGS

7.1 Key Findings

- The toolkit provides a simple and explainable approach to identifying common phishing indicators.
- The rule-based scoring model can classify sample messages into Low, Medium, and High Risk categories.
- Testing demonstrates that messages containing multiple suspicious indicators receive higher risk classifications.
- The awareness guide provides practical steps for identifying and responding to suspicious communications.
- Safe synthetic sample messages allow the toolkit to be tested without using real credentials or confidential information.

7.2 Overall Findings

Overall, the project demonstrates that a basic rule-based approach can be used to introduce users to phishing detection and cybersecurity awareness. The combination of indicator detection, risk scoring, sample message testing, and awareness guidance provides an understandable workflow for evaluating suspicious communications. The project also highlights that rule-based detection has limitations and should be improved with broader datasets and more advanced techniques for practical real-world use.

8. CHALLENGES AND LIMITATIONS

8.1 Challenges

- Identifying common phishing indicators using simple rule-based checks.
- Designing suitable risk scores for different types of suspicious indicators.
- Preparing safe and synthetic sample messages for testing.
- Avoiding false positives when legitimate messages contain suspicious-looking words.
- Ensuring that the detection logic remains simple and understandable.
- Testing different message patterns to verify consistent risk classification.

8.2 Project Limitations

The project has some limitations because the phishing detection toolkit uses predefined rules and basic heuristic checks. It does not use machine learning, real-time threat intelligence, domain reputation services, or live phishing databases. Therefore, advanced or highly customized phishing attempts may not always be detected accurately. The toolkit is designed for educational and awareness purposes and should not be considered a replacement for professional cybersecurity solutions.

8.3 Safety and Ethical Considerations

The toolkit is developed strictly for educational and awareness purposes. All testing is performed using safe, synthetic sample messages and non-sensitive information. No real user credentials, private data, confidential information, or unauthorized systems are used during the project. The phishing examples are intended only to demonstrate common warning signs and should not be used for malicious activities.

9. FUTURE SCOPE / IMPROVEMENTS

9.1 Future Improvements

- Integration of machine learning techniques for improved phishing detection.
- Use of real-time threat intelligence and phishing databases.
- Advanced URL and domain reputation analysis.
- Detection of suspicious email attachments and file types.
- Development of a graphical user interface for easier use.
- Addition of multilingual phishing detection capabilities.
- Integration with email security and awareness platforms.
- Improved reporting and analytics for phishing detection results.

9.2 Scalability and Environment

The toolkit can be further enhanced to support larger datasets and more advanced phishing detection requirements. The system can be integrated with centralized security awareness platforms and email security solutions. Additional detection rules, threat intelligence sources, machine learning models, and improved reporting features can be added as the project grows. These enhancements can make the toolkit more scalable, accurate, and useful for practical cybersecurity awareness and training.

10. CONCLUSION

The Cybersecurity Awareness and Phishing Detection Toolkit provides a simple and practical approach to improving users' understanding of phishing threats. The project combines phishing awareness, rule-based email analysis, risk classification, sample phishing scenarios, and testing activities to demonstrate how suspicious messages can be identified.

The testing process involved 20 safe sample messages and produced different risk classifications based on predefined indicators. The results demonstrate that the toolkit can identify common phishing characteristics and provide understandable risk levels and safety recommendations. Although the system has limitations, it provides a useful educational foundation for developing basic phishing detection and cybersecurity awareness skills.

11. REFERENCES

1. **National Institute of Standards and Technology (NIST).** *Cybersecurity Framework (CSF) 2.0.*
2. **Cybersecurity and Infrastructure Security Agency (CISA).** *Recognize and Report Phishing.*
3. **OWASP Foundation.** *OWASP Phishing Guidance and Security Awareness Resources.*
4. **Python Software Foundation.** *Python Documentation.*
5. **National Institute of Standards and Technology (NIST).** *Phishing and Social Engineering Awareness Resources.*